

Mike Caie on ‘Being Hesperus and Being Phosphorus’

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1 Frege’s Puzzle with names versus Frege’s Puzzle with predicates

(1) seems like it might be true (on a univocal reading):

- (1) Sam believes that Phil is a woodchuck, but Sam does not believe that Phil is a groundhog

But sentences like (2) also seem true, and are *prima facie* inconsistent with (1):

- (2) a. The property of being a woodchuck is the property of being a groundhog
b. Being a woodchuck is being a groundhog
c. To be a woodchuck is to be a groundhog
d. The woodchuck is the groundhog.

MC’s claim: the inconsistency is genuine, at least in the case of (2a). And we should react to it by keeping (1) and rejecting (2a) (and maybe also (2b–d)?).

However one ‘can tell a story about what unites such distinct but apparently identical properties’: they are *metaphysically equivalent*, i.e. metaphysically necessarily coextensive.

- A nearby alternative to MC’s view would analyse (2a–d) as true predications of metaphysical equivalence, rather than false predications of “identity in the logical sense”. This would give a Leibniz’s-Law-rejecting view of “identity in the ordinary sense” (for properties), à la Caie, Goodman, and Lederman 2020.

One goal of the paper is to propose an analysis of metaphysical necessity (in terms of counterfactuals) that makes sense of this, in a setting where there is also a *broadest* necessity, such that distinct properties cannot be broadly necessarily coextensive. (As a bonus, this theory also defines a notion of a “base” proposition, property, or relation (PRP), such that within any class of metaphysically equivalent PRPs, there is exactly one base PRP.)

The main topic of the paper is parallel puzzles involving *proper names*, such as the inconsistency between (3) and (4)?

- (3) Sam believes that Hesperus is visible, but Sam does not believe that Phosphorus is visible.
(4) Hesperus is Phosphorus.

Given his treatment of (1), one might expect that MC would likewise deny (4), and perhaps propose some fallback relation, weaker than identity (in the logical sense), as holding between Hesperus and Phosphorus. But there are problems

with this route. For one thing it’s just hard to believe. For another: what could this fallback relation be? There is no natural candidate defined just in terms of metaphysical necessity.

Instead, MC defends a *descriptivist* treatment of names like ‘Hesperus’ and ‘Phosphorus’.

- Following Montague (1974), proper names are treated as having semantic values of the same type as those of quantifiers like ‘every dog’: *property classifiers*, things of type $(e \rightarrow t) \rightarrow t$.
- Departing from Montague, the property classifiers expressed by proper names are *not* required to be “lifts” of individuals, i.e. property classifiers of the form $\lambda X.Xa$.
 - Thus, ‘There is no individual such that for Hesperus to be visible is for it to be visible’ is true, just as ‘There is no individual such that for every planet to be visible is for it to be visible’ is true.
- Following Russell-style descriptivism, these property classifiers are nevertheless taken to be derivable from underlying *properties* of individuals. They are of the form $\lambda X.\exists y.Fy \wedge \forall z.(Fz \rightarrow (z = y \wedge Xz))$, where F is the underlying property.
 - Note that when Q is definable from F in this way, F is identical to $\lambda x.Q(\lambda y.y = x)$.
- Departing from Russell, the underlying properties in question are taken to be *singular* properties: i.e., properties which *broadly necessarily have exactly one instance*.
 - Note that when F is such a property, ‘some F ’, ‘every F ’, and Russellian ‘the F ’ are equivalent: $(\lambda X.\exists y.Fy \wedge Xy) = (\lambda X.\forall y.Fy \rightarrow Xy) = (\lambda X.\exists y.Fy \wedge \forall z.(Fz \rightarrow z = y \wedge Xz))$.
- These underlying singular properties are metaphysically (though not broadly) equivalent to *haecceities*: properties of the form $(\lambda x.x = a)$.
 - This implies that the property classifiers are metaphysically (though not broadly) equivalent to Montague lifts.

2 A dialectical complication related to logical omniscience

For most of the paper, MC is assuming that ‘believe’ takes a type- t (propositional) argument, meaning that broadly necessarily equivalent sentences can be substituted *salva veritate* under ‘believe’. But the last section sketches of a treatment of logical omniscience that would give this up, allowing for example that (in some contexts), (5a) is true without (5b) being true:

- (5) a. Sam believes that Venus is visible and Venus is not visible.
- b. Sam believes that there is something that is not identical to itself.

How could this be achieved? MC mentions a possible view that posits hidden variables in the prejacent of ‘believe’ (e.g. next to the occurrences of ‘Venus’ in (5a)), which get existentially quantified by ‘believe’ (so the argument of ‘believe’ is really of type $\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t$, where $\sigma_1 \dots \sigma_n$ are the types of these hidden variables).

On any view like this, attitude reports do not in general license substitution of [what would outside of an attitude report be] co-denoting expression.

They might nevertheless license such substitution when the expressions in question are both *simple*; that would still be enough to drive the argument against (2) and against the claim that ‘Hesperus’ and ‘Phosphorus’ denote the Montague lifts.

However, this simplicity-restricted substitution of co-denoting expressions doesn’t seem so well-motivated as to be a solid foundation for drawing major metaphysical conclusions (such as the failure of physicalist supervenience theses for the broadest necessity). One could imagine views more like Frege’s, where pairs like ‘Hesperus’/‘Phosphorus’ or ‘woodchuck’/‘groundhog’ differ only in some special semantic dimension that only comes into play inside attitude reports.

3 What sorts of property-classifiers do proper names express?

MC argues, appealing to judgments about examples, that any property classifier Q expressed by a proper name is:

- (i) Negation scopeless: $\Box_B \forall P^{e \rightarrow t} (\neg Q(P) \leftrightarrow Q(\lambda x^e \neg Px))$.
- (ii) Upwards monotonic: $\Box_B \forall P_1 P_2 (QP_1 \wedge P_1 \leq P_2 \rightarrow QP_2)$.
- (iii) Closed under conjunction: $\Box_B \forall P_1 P_2 (QP_1 \wedge QP_2 \rightarrow Q(\lambda x. P_1 x \wedge P_2 x))$, plus an infinitary extension.
- (iv) Applicatively coherent: $\Box_B \forall x \forall P (Q(\lambda y. y = x) \rightarrow (QP \leftrightarrow Px))$. (I think that in the presence of the other properties, this is equivalent to $\Box_B \forall p (Q(\lambda y. p) \rightarrow p)$.)

The conjunction of these turns out to be equivalent to being derived from a *singular property* in the way explained above.

The most controversial of these is negation scopelessness, which MC argues for from the seeming equivalence of (6a) and (6b):

- (6) a. Hesperus isn’t visible.
- b. It isn’t the case that Hesperus is visible.

This is a little quick: if the denotation of ‘Hesperus’ weren’t negation-scopeless, we might expect (6a) to be scope-ambiguous in the same way as ‘Everyone isn’t

here yet’, with one reading equivalent to (6b). We could avoid this issue by replacing (6a) with something like

- (7) Hesperus is such as to not be visible.

“Orthodox” descriptivists (as well as “negative contingentists” like Stalnaker (1994)) would posit a big contrast here: if Hesperus hadn’t existed, the proposition expressed by (6b) would have been true while the one expressed by (7) wouldn’t.

I agree that there’s a good case for negation scopelessness. But if anything, I find the case here to be weaker than the case for other scopelessness claims that MC would reject, such as with respect to attitude verbs:

- (8) ? Hesperus is believed to be visible by Sam, but Sam doesn’t believe that Hesperus is visible

epistemic modals:

- (9) ? Hesperus is a planet, and it might be that you see Hesperus tonight, but Hesperus is not a planet that you might see tonight

counterfactuals (whatever the modal status of the antecedent):

- (10) If Joe’s theory were true, Mars would have been covered with water in the past; but Mars is not such that if Joe’s theory were true, it would have been covered with water in the past

higher order identities:

- (11) ? To be in the Tri-State Area is to be in NY, NJ, or CT, but the Tri-State Area is not such that to be in it is to be in NY, NJ, or CT

or random higher order predications:

- (12) ? Visiting Hesperus is on my bucket list, but Hesperus is not such that visiting it is on my bucket list.

MC does think that proper names are scopeless with respect to counterfactuals with *metaphysically possible* antecedents, taking this to have been Kripke’s key insight. But I think this misses the generality of the linguistic phenomenon.

I would advocate that the view that names are scopeless *simpliciter*: a sentence $\phi[N]$ containing a name N is equivalent to ‘ N is such that $\phi[it]$ ’.

4 Caie’s counterfactual theory of metaphysical necessity

MC wants to give an account of *metaphysical necessity* that makes it ‘theoretically important’ enough for claims of metaphysically equivalence to have some chance of serving as adequate substitutes for higher order identities like those in (2). His strategy is to explain it in terms of *counterfactual conditionals*.

- A *prima facie* worry about this general strategy: counterfactuals are widely agreed to be highly vague and context-sensitive, and one might naturally worry that other notions defined in terms of them would inherit this. Of course, one can posit that the differences between interpretations of the counterfactual wash out when we plug them into the relevant definition. But that sounds hard to justify.
- Another *prima facie* worry: counterfactual conditionals are intimately related to *indicative* conditionals, but the definitions don't have much plausibility if the conditionals are interpreted as indicative. How plausible is it that the special effects of counterfactual marking always impose these sharp constraints on the interpretation of conditionals?

A simple account of this form would define '*p* is metaphysically necessary' with '*p* would be true no matter what else was true', or (equivalently) '*p* would be true even if it weren't'.

- MC rejects this because he thinks there are false counterfactuals with metaphysically impossible antecedents, such as
- (13) If Hesperus were visible and Phosphorus not, then Hesperus would still be Phosphorus.
- A different reason to reject it would be the claim that there are vacuously true counterfactuals with metaphysically *possible* antecedents. This is something I would want to say (in many contexts), since I think the inference from 'it couldn't be that *P* and not *Q*' to 'if it were that *P*, it would be that *Q*' is valid (where 'could' is often much narrower than metaphysical possibility). This view also makes trouble for MC's final analysis.

MC's clever strategy is built around the claim that metaphysical possibility is a status *preserved* under counterfactuals.

Possibility-preservation (It's broadly necessary that:) whenever $p > q$ and it's metaphysically possible that *p*, it's metaphysically possible that *q*.

Metaphysical necessity isn't the only status that works like this: broad necessity does, and one might think that (e.g.) physical necessity does too. MC proposes to define metaphysical necessity as the *second-broadest* S5 modality that obeys Possibility-preservation.

Why think that there *is* a second-broadest, or that if there is one, it applies to such propositions as that Hesperus is Phosphorus? MC develops an interesting and specific theory of the structure of the counterfactual orderings which entail this. But ultimately, this theory is supported on abductive grounds, and I ended up not sure how much of an explanatory advance it makes over a theory that simply posits that there's a second-broadest.

Some doubts about Possibility-preservation:

- (14) a. If business had gone on as usual at the *Daily Planet* despite the presence of Kryptonite, we would have learnt that Clark Kent wasn't Superman.

- b. If none of the gases obtained from water by electrolysis had been combustible, it would have been established that water was not H₂O.

Given the factivity of 'learn' and 'establish' and the metaphysical necessity of identity, the consequents must be metaphysically impossible; but the antecedents don't seem to be.

5 Some questions about metasemantics

There's a well-known view about 'theoretical terms' (Lewis 1970) according to which, when we have established that a certain role is uniquely occupied, it's straightforward to introduce a new word that denotes the unique occupant of the role. Given this view, we could easily introduce new words that express *base* properties and relations—e.g. we could say 'Let 'positron' express the unique base property that plays such-and-such physical role', or 'Let 'Neptune' express the Montague lift of the planet that is perturbing the orbit of Uranus'.

Does MC think that this just doesn't work?

- If not, I suppose the relevant terms only end up denoting things *metaphysically equivalent* to the occupants of the roles. Can anything be said about which such things get selected?
- I think MC would grant that certain *logical constants*, such as $=_e$, denote base entities. Is there a story about how these managed to scrape past the filter that generally blocks such denotations?
- Lots of words, e.g. the counterfactual conditional, have denotations that play certain rich "logical" roles with broad necessity. But if our use of a word can constrain its denotation in this way, why can't it do so in the ways that would allow forcing a base denotation?
- The picture seems to be one where when words are introduced separately, they not only express *distinct* things, but express things that are in some sense *independent* (according to the broadest necessity). E.g. it's broadly possible that there are woodchucks but no groundhogs, or vice versa. Is there a general principle underlying this?

References

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